SGT Black Hole Physics: The Complete Picture from Static Structure to Rotational Corrections

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Abstract

The Spatial-Pressure Gravitational Theory (SGT) provides a complete picture of black hole physics that is fundamentally distinct from General Relativity (GR). This paper systematically elaborates the core results of SGT black hole physics across three progressive tiers. **Tier I**: In the spherically symmetric background, the modulation of the effective gravitational constant by the medium order parameter $\chi(r)$ ($G_{\text{eff}} = G \cdot \chi$) causes the event horizon to contract to $r_H = 1.668M$ and the Innermost Stable Circular Orbit (ISCO) to shift inward to $4.23M$ (GR: $6.00M$). **Tier II**: Under metric perturbations, the ISCO exhibits an ε^2-order rigid response (the first-order correction strictly vanishes), the photon sphere exhibits an ε^1-order soft response, and the event horizon exhibits an ε^0-order invariant response—this three-tier rigidity hierarchy originates from the differences in the competition mechanisms between gravity and centrifugal force in the effective potentials of different structures. **Tier III**: In the rotating background, this paper proposes the **Local Effective Mass Approximation (LEMA)**, introducing the G_{eff} modulation into the effective potential equation for the rotating ISCO for the first time. The study reveals a unique self-regulating negative feedback mechanism in SGT: rotation drives the ISCO inward, but the inward shift increases $\chi(r)$ and restores the effective gravity, partially counteracting the rotational effect, resulting in an 86% suppression of the first-order rotational response of the ISCO ($C_1^{\mathrm{SGT}} = -0.466$ vs $C_1^{\mathrm{Kerr}} = -3.266$). The findings across all three tiers point to the same physical root—the non-emptiability of the spatial-energy medium ($\chi \geq 0$)—and are all independently testable by current and next-generation astronomical observations. As the definitive paper on SGT black hole physics, this work integrates spherically symmetric black holes, structural stability analysis, and rotational corrections, constructing a complete theoretical picture from static to rotating regimes.

Keywords: Spatial-Pressure Gravitational Theory; Black Hole Physics; ISCO Shift; Structural Rigidity; Local Effective Mass Approximation; Rotational Negative Feedback

I. Introduction

The black hole physics of General Relativity (GR) has achieved brilliant observational verification in the weak-to-medium field strength regimes. The 2019 Event Horizon Telescope (EHT) imaging of the M87* black hole shadow revealed a ring-like structure whose size and shape are consistent with the predictions of the GR Kerr metric. The electromagnetic counterpart observation of the 2017 GW170817 gravitational wave event constrained the relative deviation between the speed of gravitational waves and the speed of light to the order of 10^{-15}. Observations of the Fe Kα line profile are also consistent with the GR ISCO predictions within current precision.

However, the black hole physics of GR exhibits three fundamental silent zones. First, the singularity problem—the Penrose-Hawking singularity theorems declare the failure of physics at singularities, yet GR itself cannot resolve them. Second, the dark matter problem—galactic rotation curves require a vast amount of invisible matter, which GR cannot explain from its own field equations. Third, the structural stability problem—GR has never systematically answered: when the metric is slightly modified, do the core structures such as the ISCO, photon sphere, and event horizon remain stable?

The Spatial-Pressure Gravitational Theory (SGT) provides a unified answer to these three problems. The core postulate of SGT is that the vacuum is not empty but filled with a continuous elastic medium—the "spatial-energy medium." Matter acts as a local defect in the medium, and gravity is the tension gradient generated when the medium is stretched. Within this framework:

1. Singularities are physically truncated by non-emptiability—the medium cannot be infinitely compressed, and the $\chi = 0$ surface forms an insurmountable dynamical barrier.
2. Galactic rotation curves naturally yield MOND-like behavior in the weak-field limit—requiring no dark matter particles.
3. The ISCO, photon sphere, and event horizon exhibit distinct stability levels under metric perturbations—this hierarchical structure is a direct consequence of the medium's existence.

Focusing on black hole physics, this paper systematically elaborates the core results of SGT across three progressive tiers: the baseline shift of static spherically symmetric black holes (Section II), the structural stability hierarchy under metric perturbations (Section III), and the G_{eff} negative feedback mechanism in rotating black holes (Section IV). The core argument of this paper can be condensed into **"one mechanism, three tiers"**: G_{eff} **Modulation Negative Feedback**—the dynamic modulation of the effective gravitational constant by the medium order parameter $\chi(r)$—is the unified physical root of SGT black hole physics. At the zeroth order ($a = 0$), it directly alters the baseline positions of the ISCO and the horizon; at the first order ($a \neq 0$), it suppresses rotational effects via negative feedback; at the second order and beyond, it endows different structures with distinct perturbative response orders.

II. SGT Spherically Symmetric Black Holes: Baseline Shifts and G_{eff} Modulation

2.1 SGT Field Equations and the *Ω*-A1 Static Solution

The field equations of SGT are derived via the variational principle of the elastic medium action. In the spherically symmetric static vacuum, the field equations reduce to the *Ω*-A1 ODE system:

$$\frac{dm}{dr} = \frac{\kappa}{2} r^2 \rho_T(f), \qquad \frac{d\nu_0}{dr} = \frac{m - \frac{\kappa}{2} U(f) r^3}{r(r-2m)}$$

where $A_0 = e^{2\nu_0}$, $B_0 = (1 - 2m/r)^{-1}$, $f = 1 - A_0$, and $\kappa = 8\pi G K$. The effective energy density of the medium is $\rho_T(f) = U(f) + 2U'(f)(1-f)$, which is bounded over the entire interval $f \in [0,1]$.

The key outputs of the *Ω*-A1 numerical solution include:

- Event horizon radius: r_H = *1.668M* (GR: $2M$, a 16.6% shrinkage).
- Photon sphere radius: r_{ph} = *2.9954M* (GR: $3M$, a 0.15% deviation).
- ISCO radius (based on standard geodesics): $r_{\text{ISCO}}^{\text{geo}} \approx 6M$ (nearly identical to GR, as the *Ω*-A1 metric recovers the Schwarzschild form for r > *3.5M*).

2.2 G_{eff} Modulation: Effective Description at the Orbital Dynamics Level

The core physical effect of SGT is not a **direct alteration** of the spacetime geometric metric, but the dynamic modulation of the gravitational coupling strength by the medium. The order parameter $\chi(f) = (1-f)/(1+f^2)$ is significantly less than 1 within the extended medium shell, causing the effective gravitational constant $G_{\text{eff}} = G \cdot \chi(r)$ to be suppressed.

Crucial Clarification: G_{eff} modulation is not "another metric" outside the exact *Ω*-A1 solution, but rather an **Effective Description at the Orbital Dynamics Level** for the same physical reality. It repackages the contribution of the medium energy density $\rho_T(f)$ to the gravitational field into a local effective mass $M_{\text{eff}}(r) = M\chi(r)$ felt by a test particle. While the metric **asymptotically reduces** to the standard Schwarzschild form in the far-field regime ($r \gg 100M$), **significant residual medium effects persist in the intermediate field** (e.g., $\chi \approx 0.6$ at the ISCO radius $r \sim 4.2M$). The G_{eff} modulation more intuitively embodies the physical picture of "medium shielding gravity," and is therefore selected as the foundational framework for ISCO calculations.

Under G_{eff} modulation, the spherically symmetric effective potential is:

$$V_{\text{eff}}^{\text{SGT}}(r, L) = \left(1 - \frac{2M\chi(r)}{r}\right)\left(1 + \frac{L^2}{r^2}\right)$$

Table 1 presents the comparison of $A_0(r)$ between the *Ω*-A1 metric and the G_{eff} modulated metric at key radii:

r/M	A_0 (**_Ω_-A1**)	A_0 (G_{eff})	$\chi(r)$	**Deviation**
3.0	0.333	0.846	0.231	+154%
4.0	0.500	0.800	0.400	+60%
5.0	0.600	0.793	0.517	+32%
6.0	0.667	0.800	0.600	+20%
10.0	0.800	0.840	0.800	+5%

Table 1: Comparison of $A_0(r)$ between the Ω-A1 metric and the G_{eff} modulated metric. Within the medium shell ($r < 5M$), the A_0 of the G_{eff} modulated metric is significantly larger than the Ω-A1 value, because $\chi(r) < 1$ makes the effective mass $M_{\text{eff}} = M\chi(r) < M$, making the gravitational field appear weaker. The two converge asymptotically in the far field.

2.3 The Origin of the ISCO Shift

Deriving the ISCO conditions $V_{\text{eff}}' = V_{\text{eff}}'' = 0$ from the G_{eff} modulated effective potential and solving numerically yields:

$$r_{\text{ISCO}}^{\text{SGT}}(a = 0) = 4.23M$$

This represents a -29.5% deviation from the GR value of $6.00M$. The physical reason is that G_{eff} is significantly suppressed within the medium shell ($\chi(4.23M) = 0.426$, meaning the effective mass is only 43% of the true mass); the gravitational field weakens, allowing orbits to remain stable at positions closer to the center. This value is fully consistent with the definition in Constitutional Baseline §2.10 and constitutes the most critical testable prediction of SGT in the realm of static black holes.

III. Gravitational Structural Stability Analysis: The Three-Tier Rigidity Hierarchy

The SGT medium not only corrects the static positions of black hole structures but also alters their stability behaviors under metric perturbations. The CH Series framework establishes a universal testing protocol for gravitational structural stability. By applying a Type-A metric perturbation $g_{tt} =- (1 - 2M/r)(1 + \varepsilon M/r)$ to the Schwarzschild background, the responses of the ISCO, photon sphere, and event horizon are systematically analyzed.

3.1 The ε^2-Order Rigidity of the ISCO

Under perturbation, the ISCO radius can be expanded as $r_{\mathrm{ISCO}}(\varepsilon) = 6M + c_1\varepsilon + c_2\varepsilon^2 + ...$ The analytical derivation in CH-0001 rigorously proves that: $c_1 = 0$. The first-order correction strictly vanishes.

The mathematical root of this conclusion is fully proven in Appendix A. The physical mechanism is as follows: the metric perturbation ε simultaneously corrects both the gravitational potential and the centrifugal potential—the gravitational potential is corrected via g_{tt}, while the centrifugal potential is corrected via the adjustment of $L^2(r, \varepsilon)$—at $r = 6M$, these two first-order effects are equal in magnitude and opposite in sign, exactly canceling each other out. The response order of the ISCO is ε^2, classified as **STRUCTURE-PRESERVED**.

3.2 The ε^1-Order Softness of the Photon Sphere

The characteristic equation for the photon sphere possesses an exact analytical solution under Type-A perturbation:

$$r_{\mathrm{ph}}(\varepsilon) = M\frac{3(2 - \varepsilon) + \sqrt{9(2 - \varepsilon)^2 + 64\varepsilon}}{4}$$

Perturbative expansion yields $r_{\mathrm{ph}}(\varepsilon) = 3M - M\varepsilon/6 + O(\varepsilon^2)$. The first-order coefficient is $c_1 =- 1/6 \neq 0$. The photon sphere is first-order sensitive to metric perturbations.

Physical reason: The photon sphere is a pure gravitational geometric effect; the null geodesics of photons depend solely on the geometric structure of g_{tt}, with no centrifugal potential involved. The perturbation directly corrects g_{tt} without any counteracting mechanism. It is classified as **STRUCTURE-DEFORMED**.

3.3 The ε^0-Order Invariance of the Event Horizon

The characteristic equation for the event horizon is $g_{tt} = 0$. Under the Type-A correction, the physical event horizon always remains at $r = 2M$, unaffected by ε. It is classified as **STRUCTURE-PRESERVED** (ε^0-order).

3.4 The Three-Tier Hierarchy and Its Physical Root

Structure	Response Order	First-order Coefficient	Physical Mechanism	Classification
Event Horizon	ε^0	0	$g_{tt} = 0$ is independent of ε	Rigid
ISCO	ε^2	0	First-order cancellation between centrifugal and gravitational potentials	Rigid
Photon Sphere	ε^1	$-1/6$	Pure gravitational geometry, no counteraction	Soft

This hierarchical structure serves as indirect verification of the existence of the SGT medium—the difference in response orders of different structures to the same perturbation stems from whether they involve the competition between centrifugal and gravitational potentials. In GR, the responses of all three structures to metric perturbations are ε^1-order, with no hierarchical difference.

IV. SGT Rotating Black Holes: Local Effective Mass Approximation and G_{eff} Negative Feedback

4.1 Self-Consistent Construction of the Rotating ISCO Equation: Local Effective Mass Approximation (LEMA)

In the Kerr background, the ISCO condition for equatorial circular orbits is given by the Bardeen formula:

$$1-\frac{3M}{r}+\frac{2a\sqrt{M}}{r^{3/2}}=0 \quad \text{(prograde)}$$

In SGT, the modulation of the medium on gravitational coupling manifests at the dynamical level of test particles. We propose the **Local Effective Mass Approximation (LEMA)**: the orbiting particle feels not the asymptotic global mass M, but the local effective mass $M_{\mathrm{eff}}(r)=M\chi(r)$ shielded by the medium. Crucially, this modulation does not alter the underlying geometry of the spacetime; **the metric tensor retains its standard Kerr form, whereas the gravitational coupling enters the effective potential of the test particle as a locally corrected variable.**

At the effective potential level, replacing the global mass M in the Bardeen condition with the local effective mass $M\chi(r)$ yields the implicit equation for the rotating ISCO in SGT:

$$F(r_*) = 1 - \frac{3M\chi(r_*)}{r_*} + \frac{2a\sqrt{M\chi(r_*)}}{r_*^{3/2}} = 0$$

Key Characteristic: $\chi(r_*)$ is a function of r_* itself—the ISCO position depends on χ, and χ depends on the ISCO position. This is a self-consistent equation that must be solved iteratively.

Derivation Property: LEMA is a phenomenological extension at the orbital dynamics effective potential level, not a result strictly derived from the complete axisymmetric SGT field equations. However, numerical verification in the range $a \leq 0.5$ (see §4.2) demonstrates high consistency with existing exact SGT solutions. **Validation Confidence Level: Grade A** ($a \leq 0.5$ numerical verification).

4.2 $C_1 = -0.466$: 86% Rotational Suppression

Numerical solutions for $a = 0, 0.1, 0.2, 0.3, 0.4, 0.5$ and polynomial fitting of the results yield the rotational expansion of the ISCO:

$$r_{\mathrm{ISCO}}^{\mathrm{SGT}}(a) = 4.23M - 0.466aM - 0.129a^2M + \ldots$$

Table 2 presents the comparison of the ISCO between SGT and GR Kerr in the range $a \leq 0.5$. The relative deviation is defined as $\Delta\% = \frac{r_{\mathrm{ISCO}}^{\mathrm{SGT}} - r_{\mathrm{ISCO}}^{\mathrm{Kerr}}}{r_{\mathrm{ISCO}}^{\mathrm{Kerr}}} \times 100\%$. The SGT ISCO is consistently smaller than the GR Kerr value, but **the relative gap narrows significantly** as spin increases (from -29.5% to -5.4%), demonstrating the strong resistance of the G_{eff} modulation against rotational inward shift.

a/M	$r_{\text{ISCO}}^{\text{SGT}}/M$	$r_{\text{ISCO}}^{\text{Kerr}}/M$	$\Delta\%$	$\chi(r_{\text{ISCO}})$	**Confidence Level**
0.0	4.230	6.000	-29.5%	0.426	A
0.1	4.183	5.669	-26.2%	0.439	A
0.2	4.137	5.329	-22.4%	0.451	A
0.3	4.092	4.979	-17.8%	0.464	A
0.4	4.048	4.615	-12.3%	0.477	A
0.5	4.005	4.233	-5.4%	0.490	A

Table 2: Comparison of the ISCO between SGT and GR Kerr ($a \leq 0.5$).

Core Finding: $C_1^{\text{SGT}} = -0.466$, which is only 14% of the GR Kerr value $C_1^{\text{Kerr}} = -3.266$—**the rotational response of the ISCO is suppressed by 86%**.

4.3 The Physical Picture of the G_{eff} Negative Feedback Mechanism

The suppression of rotation in SGT originates from a unique **Self-regulating Negative Feedback Loop**:

1. **Rotation drives inward shift**: As in Kerr, rotation moves the ISCO toward the center ($\partial r_{\text{ISCO}}/\partial a < 0$).
2. **Inward shift enters the high-χ region**: ISCO shifts inward → r_{ISCO} decreases → enters deeper into the medium shell → $\chi(r_{\text{ISCO}})$ increases ($\partial\chi/\partial r < 0$).
3. **Effective gravity restores**: χ increases → $M_{\text{eff}} = M\chi$ increases →

effective gravity strengthens → the orbit must move further out to remain stable.

4. **Negative feedback closes**: The enhancement of effective gravity directly resists the inward shift caused by rotation.

Mathematical formulation: At the ISCO, the inward shift corresponds to $\frac{\partial r_{\mathrm{ISCO}}}{\partial a} < 0$, and entering the high-$\chi$ region implies $\frac{\partial \chi}{\partial r} < 0$. The chain rule yields:

$$\frac{\partial \chi}{\partial a} = \frac{\partial \chi}{\partial r_{\mathrm{ISCO}}} \cdot \frac{\partial r_{\mathrm{ISCO}}}{\partial a} = (-) \cdot (-) > 0$$

This positive derivative is the mathematical essence of the negative feedback: the increase in spin a indirectly increases the local effective mass χ, thereby enhancing gravity and resisting further inward shift. The negative feedback suppresses the inward shift rate of the ISCO (C_1 is suppressed by 86%).

In GR, M is a constant, and this feedback loop does not exist. In SGT, $M_{\mathrm{eff}} = M\chi(r)$ is dynamically changing, and the variation of $\chi(r)$ with r automatically generates a restoring force that resists rotation. This is the core dynamical feature that distinguishes SGT from GR (and all constant-coupling modified gravity theories).

Table 3 presents the effective mass analysis of the ISCO positions, intuitively demonstrating the strength of the negative feedback:

a/M	r_{ISCO}/M	$\chi(r_{\mathrm{ISCO}})$	M_{eff}/M	$r_{\mathrm{ISCO}}/M_{\mathrm{eff}}$
0.0	4.230	0.426	0.426	9.93
0.1	4.183	0.439	0.439	9.53
0.2	4.137	0.451	0.451	9.17

0.3	4.092	0.464	0.464	8.82
0.4	4.048	0.477	0.477	8.49
0.5	4.005	0.490	0.490	8.18

Table 3: Effective mass analysis of ISCO positions. The ratio $r_{\text{ISCO}}/M_{\text{eff}}$ in SGT is always greater than the GR Kerr value of *6.00*, indicating that even as the ISCO shifts inward, the effective gravity remains weaker than in GR—this is the continuous resistance of G_{eff} modulation against rotational effects.

4.4 Non-Emptiability Surface, Quadrupole Moment, and Maximum Spin

(This section is based on the physical reasoning and order-of-magnitude estimations from Report 022.)

In the rotating background, the $\chi = 0$ surface (the non-emptiability limit surface) changes from spherical to oblate spheroidal. Based on a simplified model of the χ field perturbation equations, the oblateness is estimated to be approximately $1.001 - 1.015$ for $a = 0.3$. **Validation Confidence Level: Grade B** (based on a simplified perturbation model, excluding full non-linear calculations).

A simplified calculation of the quadrupole moment—based on the geometric part of the ODE, temporarily excluding the medium source term—yields $Q^{\text{SGT}}/Q^{\text{Kerr}} \approx 0.58$ (-42% deviation). **Validation Confidence Level: Grade**

B (based on the geometric part of the ODE; the medium source term is pending).

The maximum spin is constrained by the elastic limit of the medium, with a comprehensive estimate based on physical reasoning yielding $a_{\max} \approx 0.90M$.

Validation Confidence Level: Grade C (based on physical reasoning, excluding full non-linear calculations).

V. Summary of Testable Predictions

All predictions of SGT black hole physics are arranged according to their testable timelines below.

Near-term (2027-2030):

Prediction	SGT Value	GR Value	Deviation	Detector	Confidence Level
Fe Kα red wing cutoff energy	~3.6 keV	~3.0 keV	+20%	XRISM	A
ISCO rotational coefficient C_1	-0.466	-3.266	-86%	Athena	A

Mid-term (2030-2040):

Prediction	SGT Value	GR Value	Deviation	Detecto r	Confidenc e Level
ISCO radius ($a = 0$)	4.23M	6.00M	-29.5%	LISA	A
ISCO-Kerr gap narrowing	$\Delta\% \approx -5.4\%$	$\Delta\% = 0\%$ (Baseline)	Gap narrows	Athena	A

($a \approx 0.5$)					
Quadrupole moment deviation	$Q/Q_{\text{Kerr}} \approx 0.58$	$Q = -a^2M$	-42%	LISA	B
Medium shell GW echoes	Delay ~0.3M	None	Qualitatively different	ET	B

Long-term (2040+):

Prediction	SGT Value	GR Value	Deviation	Detector	Confidence Level
Maximum spin limit	$a_{\max} \approx 0.8 - 0.99M$	$a = 1$ reachable	Qualitatively different	ngEHT	C

Unique SGT Predictions (GR cannot provide these at all):

1. 86% suppression of ISCO rotation (G_{eff} negative feedback): LISA (2035+).
2. Significant narrowing of the ISCO-Kerr gap at high spins: Athena.
3. Medium shell gravitational wave echoes ($\chi = 0$ surface reflection): ET.
4. Existence of the non-emptiability surface ($\chi \geq 0$ constraint): Echo reflection coefficient.

VI. Honest Annotations and Theoretical Boundaries

Deterministic Conclusions (Grade A Evidence):

- ISCO = 4.23M ($a = 0$), a -29.5% deviation from GR.
- ISCO rotational correction $C_1 = -0.466$, suppressed by 86%.

- The perturbative rigidity hierarchy of the three structures.
- Numerical solutions for the ISCO at $a \leq 0.5$.

Estimative Conclusions (Grade B Evidence):

- Oblateness of the non-emptiability surface $1.001 - 1.015$ ($a = 0.3$).
- Quadrupole moment deviation of −42%.
- Extrapolation of the ISCO for $a = 0.5 - 0.8$.

Speculative Conclusions (Grade C Evidence):

- Maximum spin $a_{\max} \approx 0.8 - 0.99M$.
- ISCO behavior for $a > 0.8$.

Key Uncertainties:

1. Microscopic foundation of G_{eff} modulation: Currently a phenomenological description, not derived from the fundamental action.
2. Exact equation for $\chi_2(r)$: Requires 2D Ricci tensor calculations.
3. High-spin non-linear effects: Requires 2+1D PDE numerical simulations.

Future Work Directions (Prioritized):

1. Microscopic derivation of G_{eff} modulation (starting from the field equations).
2. Exact calculation of $\chi_2(r)$ (2D Ricci tensor).
3. High-spin non-linear calculations (2+1D PDE).
4. Gravitational wave echo waveform templates (the most unique prediction of SGT).

VII. Conclusion

This paper constructs a complete picture of SGT black hole physics across three progressive tiers. In the spherically symmetric static background, G_{eff} modulation causes the ISCO to shift inward to 4.23M and the event horizon to contract to 1.668M. Under metric perturbations, the ISCO, photon sphere, and event horizon exhibit a three-tier rigidity hierarchy of $\varepsilon^2 > \varepsilon^1 > \varepsilon^0$—this hierarchical structure is indirect verification of the medium's existence. In the rotating background, through the Local Effective Mass Approximation (LEMA), it is discovered that G_{eff} modulation generates a self-regulating negative feedback mechanism, suppressing the rotational response of the ISCO by 86%—this is the core dynamical feature that distinguishes SGT from GR and all constant-coupling modified gravity theories.

The findings across all three tiers point to the same physical root—the non-emptiability of the spatial-energy medium ($\chi \geq 0$)—and are all independently testable by current and next-generation astronomical observations. The Fe Kα line red wing shift (XRISM, 2027+) and the suppression of ISCO rotation (Athena, 2030s) are the two predictions with the shortest timelines, and will directly face the test of observational data within the next decade.

Ownership and Evidence Declaration

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References

[1] Bardeen, J. M., Press, W. H., & Teukolsky, S. A. (1972). Rotating Black Holes: Locally Nonrotating Frames, Energy Extraction, and Scalar Synchrotron Radiation. *ApJ*, 178, 347.

[2] Li, Z. SGT-PAPER-012: Flagship Overview Paper of Spatial-Pressure Gravitational Theory (SGT), 2026.

[3] Li, Z. SGT-PAPER-016: Theoretical Framework for Gravitational Structural Stability Testing via the CH Series, 2026.

[4] Li, Z. SGT-TH-2026-018: CH Series Gravitational Structural Stability Testing Framework, 2026.

[5] Li, Z. SGT-TH-2026-021: Technical Report on the Conceptual Closure of SGT Rotating Black Holes, 2026.

[6] Li, Z. SGT-TH-2026-022: Technical Report on Higher-Order Closure and Final Verification of SGT Rotating Black Holes, 2026.

Appendix A: Rigorous Mathematical Proof of the Vanishing First-Order ISCO Correction

(The complete proof is provided in Appendix A of SGT-PAPER-016; detailed derivations are omitted here, and only the conclusion is stated.)

Under the Type-A metric perturbation $A(r) = (1 - 2M/r)(1 + \alpha M/r)$, the ISCO characteristic equation is $F(r, \alpha) = 0$. Applying the Implicit Function Theorem, the first-order coefficient is $c_1 = -\left(\partial F/\partial\alpha\right)/\left(\partial F/\partial r\right)|_{r=6M,\alpha=0}$. Substituting $A(r, \alpha) = A_0(r) + \alpha A_1(r)$ into the explicit expression for $F(r, \alpha)$ and calculating the partial derivatives term by term, utilizing the key fact that $A_1''(6M) = 0$, it is rigorously proven that $\partial F/\partial\alpha|_{r=6M,\alpha=0} = 0$ and $\partial F/\partial r|_{r=6M,\alpha=0} \neq 0$, hence $c_1 = 0$. ■

Appendix B: Interpolation Data for $\chi(r)$

r/M	$\chi(r)$
1.668	0.000
3.0	0.231
4.0	0.400
4.23	0.426
5.0	0.517
6.0	0.600
10.0	0.800
20.0	0.952
50.0	0.992
100.0	0.998

This paper is a core publication of the SGT theoretical system. Version V1.3 (Final Audited Version), June 4, 2026. Constitutional Baseline V3.10.0. This *is a public snapshot version, intended solely for academic disclosure and timestamp priority filing; it does not contain detailed mathematical derivations.*